

Before the
LIBRARY OF CONGRESS
UNITED STATES COPYRIGHT OFFICE

In the Matter of the Copyright Office's Software-Enabled Consumer Products Study

Docket No. COLC-2015-0011

Comment of the Specialty Equipment Market Association (SEMA)

Submitted By

Specialty Equipment Market Association (SEMA)
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Dear Sir/Madam:

The Specialty Equipment Market Association (SEMA) was founded in 1963 and represents the \$36 billion specialty equipment automotive aftermarket industry of 6,633 member-companies. SEMA's members offer custom auto accessories that enhance a vehicle's appearance, safety, performance, comfort and convenience. These aftermarket products include such accessories as custom wheels and tires, turbochargers, lighting equipment, performance exhausts, suspensions, grille guards, and sunroofs, and enable hobbyists to customize their vehicles according to their own tastes and preferences. The specialty automotive aftermarket supports these hobbyists and represents a longstanding history of innovation and entrepreneurship, developing cutting-edge products that increasingly become mainstream in the auto industry.

SEMA respectfully submits comments for the Software-Enabled Consumer Products Study being undertaken by the U.S. Copyright Office in response to a request made by Senators Charles E. Grassley and Patrick Leahy of the Senate Committee on the Judiciary. The Senators requested information on five different topics, including "the provisions of the copyright law that are implicated by the ubiquity of copyrighted software in everyday products" and "whether, and to what extent, legitimate interests or business models for copyright owners and users could be undermined or improved by changes to the copyright law in this area." In considering changes to the copyright laws, SEMA supports any changes that would have the effect of improving the ability to access and utilize elements of vehicle software. To maintain a competitive marketplace for aftermarket parts, the ability to access and utilize these elements should be extended to the owner of the vehicle, shops undertaking modifications on behalf of the owner and companies developing interoperative software and vehicle components that must interact with the software. SEMA's position is clear. In any changes to the copyright laws, it is important that aftermarket companies continue to have access and the ability to reverse engineer the software of modern vehicles to achieve different functionality and interoperability with aftermarket components.

Provisions of copyright law implicated by ubiquity of copyrighted software in everyday products

Fair Use – Section 107 of the Copyright Act

Congress sought to limit the exclusive rights granted to the copyright owner by providing an exception for fair use. The concept of fair use in the Copyright Act is central to the use of copyright-protected software that runs modern vehicles.¹ In analyzing whether a particular use is a “fair use,” a court is directed to consider the use in light of four statutory factors: (1) the purpose and character of the use, (2) the nature of the copyrighted work, (3) the amount of the work used in relation to the work as a whole, and (4) the effect the use has on the market for or value of the work.

The first fair use factor asks whether the use of a copyrighted work “merely supersedes the objects of the original creation, or instead adds something new.”² Following these principles, specifically in the context of computer software, it is clear that a modifier’s use of software to allow for the installation and interoperability of specialty parts and accessories is a “legitimate purpose” that weighs in favor of fair use.

The nature of the copyrighted work weighs heavily in favor of fair use when the user is only interested in the functionality of the code, not its expression. In the recent case of *Oracle Am., Inc. v. Google Inc.*, the Federal Circuit reiterated that software, even where it contains functional elements, is entitled to copyright protection. However, the court went on to explain that “where the nature of the work is such that purely functional elements exist in the work and it is necessary to copy the expressive elements in order to perform those functions, consideration of this second factor arguably supports a finding that the use is fair.”³

The third factor only weighs against fair use if the amount used is unreasonable in light of the purpose for which the copyrighted work is used.⁴ Thus, even where the entire copyrighted work has been copied, such a finding will not defeat fair use if the copying was necessary to accomplish a legitimate purpose.⁵ The use of vehicle software to achieve aftermarket personalization, modification or other improvements supports a finding of fair use under the third factor, where the modifiers use no more than is necessary to research and ultimately implement the desired modification. Ninth Circuit authority in both *Sega v. Accolade* and *Sony v. Connectix* establishes that the copying of an entire software program may be fair use if the copying was necessary to understand the functional elements of the software.⁶ In the case of reverse engineering vehicle software, analyzing the entire software program may be critical to understand the functionality of the vehicle and, in addition, to determine how much storage is

¹ 17 U.S.C. § 107 (2016).

² *Campbell v. Acuff-Rose Music, Inc.*, 510 U.S. 569, 579 (1994).

³ *Oracle Am., Inc. v. Google Inc.*, 750 F.3d 1339, 1375 (Fed. Cir. 2014).

⁴ *Campbell*, 510 U.S. at 586-87.

⁵ See *Authors Guild, Inc. v. HathiTrust*, 755 F.3d 87, 98 (2d Cir. 2014) (finding that “[f]or some purposes, it may be necessary to copy the entire copyrighted work, in which case Factor Three does not weigh against a finding of fair use”).

⁶ *Sega Enterprises Ltd. v. Accolade, Inc.*, 977 F.2d 1510, 1526-27 (9th Cir. 1992); *Sony Computer Entertainment, Inc. v. Connectix Corp.*, 203 F.3d 596, 606 (9th Cir. 2000).

available to support additional functionality. Importantly, access to the entire work is necessary to ensure that modifications in one part of the code will not negatively impact other functionality.

The last factor also weighs in favor of a finding of fair use where the copyrighted material at issue is software controlling a motor vehicle. The market for the software does not exist independent of the new vehicle. Any aftermarket component designed to interoperate with the software will not supplant the market for the new vehicle and its original software, and purchasing the new vehicle is a prerequisite for the enthusiast interested in utilizing the aftermarket components. In fact, the ability of the secondary market to provide and enable post-sale modifications that increase a vehicle's interoperability with improved components only increases the attractiveness of that vehicle to buyers, thereby benefiting the market for the vehicle and any copyrighted works contained therein.

DMCA Exemption for Reverse Engineering

A second statutory provision that indicates Congress understood the functional nature of software and the need for other programs to interact with it can be found within the Digital Millennium Copyright Act (DMCA). The DMCA expanded copyright law to prevent circumvention of access controls put in place by a copyright owner. The intent was to address the problem of pirating traditional copyrighted works, such as photos, music and movies, using digital media. The language prohibits circumventing access controls on copyrighted works, but numerous exemptions were included to ensure the prohibition did not inhibit legitimate activities. One of the permanent exemptions written into the DMCA allows circumvention of access-control mechanisms if done for the purpose of reverse engineering to achieve interoperability with a computer program. The goal is to achieve interoperability, defined by the DMCA as "the ability of computer programs to exchange information, and of such programs mutually to use the information which has been exchanged."⁷

Use of computer code now transcends the realm of the personal computer. Through the evolution of the "internet of things," software now encompasses code that takes data from connected sensors and interprets and reacts to that information. Examples from the internet of things abound, from medical devices to vehicle-to-vehicle communications. In addressing the internet of things, regulators have expressed support for technological advancements along with concerns over the security and safety of these devices and communication pathways.⁸ Copyright issues are often not the focus of concern, but overbroad copyright protection could result in a chilling effect on innovation in the secondary market for software and products that interact with connected devices. Changes to the law that would increase the ability to rely on the fair use doctrine in situations where the copyrighted materials at issue are functional features of software controlling consumer products could help limit the chilling effect on innovation. Possible legislative solutions could include heightened pleading requirements for copyright infringement

⁷ 17 U.S.C. § 1201(f)(4) (2016).

⁸ See *The Internet of Things: FDA Releases Guidance on Securing Wireless Medical Devices – What Medical Device Manufacturers Should Know*, INFOLAWGROUP LLP (Oct. 2, 2013), www.infolawgroup.com/2013/10/articles/information-security/the-internet-of-things-fda-releases-guidance-on-securing-wireless-medical-devices-what-medical-device-manufacturers-should-know/; see also *Transportation Sec. Foxx Announces Steps to Accelerate Road Safety Innovation*, NHTSA, DOT 49-15 (May 13, 2015), <http://www.nhtsa.gov/About+NHTSA/Press+Releases/2015/nhtsa-will-accelerate-v2v-efforts>.

claims involving software-enabled consumer products. Another option would be to require a copyright owner to perform a subjective good faith analysis to determine whether the conduct at issue is for a purpose that constitutes fair use before using enforcing the DMCA's anti-circumvention provision. The Ninth Circuit recently imposed such a requirement on copyright owners using the DMCA's notice and takedown provision.⁹

Legitimate interests or business models for copyright owners and users that could be undermined or improved by changes to the copyright law in this area

Achieving interoperability of aftermarket components with vehicle software is a legitimate activity that could be impacted by the use of copyright law to limit access and use of the software, particularly the functional elements of vehicle software. Although in decades past hobbyists could install these new features mechanically, the advent of government-mandated electronic-based vehicle systems—including onboard diagnostics, tire pressure monitoring systems, and electronic stability control—has given rise to a need to more fully understand vehicle software to ensure new innovations will be compatible with existing systems.

Modern vehicles rely heavily on a network of electronic systems to collect and use data from various sensors. Sensor data is collected and processed by the vehicle's different control modules (Electronic Control Units – “ECUs” – made with microcontroller chips). Modern vehicles can have up to 70-80 ECUs, ranging from the Engine Control Module (ECM) to the Telematic Control Unit (TCU). Microcontroller chips in each module, including the ECM, store software code in the form of flash memory or Programmable Read Only Memory (PROM) that respond to the sensor data. Aftermarket companies typically replace or recalibrate microcontroller chips in the ECM for improved performance.

The California Air Resources Board (CARB) regulates changes to the ECM. Hundreds of Executive Order numbers have been issued to aftermarket companies for various changes to ECUs to improve engine performance or allow other components to be installed on a vehicle.¹⁰ For example, aftermarket turbochargers can be installed on a vehicle to increase engine efficiency, but the vehicle's ECM will need to be replaced or recalibrated to account for the change in fuel/air entering the engine.

In undertaking the Software-Enabled Consumer Products Study, please feel free to reach out to SEMA for further information on the use of vehicle software in the automotive aftermarket.

⁹ *Lenz v. Universal Music Corp.*, No. 13-606 (9th Cir. 2015) (copyright owner must perform a subjective good faith fair use analysis before issuing a DMCA takedown notice).

¹⁰ *Aftermarket Parts Database of Executive Orders*, CALIFORNIA AIR RESOURCES BOARD, <http://www.arb.ca.gov/msprog/aftermkt/devices/amquery.php> (last visited Feb. 16, 2016).